

Generative AI for Personalized Learning Experiences in Education

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Abstract

When AI is used in schools, it should help kids to learn more. Individualized learning is an idea that has been around for a while, but you could not use it until not long ago. This research shows easy, cheap, and long-lasting ways to make learning tools. The teacher's learning goals for that class were used to make our tool. This tool can be used with a learning management system already set up at an engineering college. The lesson plans had three styles. For every question, there was more than one short answer and choice. There are three ways to get each lesson. Twenty software engineering students from a well-known European school helped test it in a small way. You can make this tool with a big language model and an application programming interface (API). All the participants participated in their own wish. These things were given to everyone in the class twice. They made a list of the things they used. They also cared about what people saw, thought, and felt, and they cared about these things over time. At school, it was also more important. People paid the most attention to the tests made immediately and looked like quizzes. Because it only looked at twenty kids, the study couldn't say what its results mean for all of them. The activation level for all tests was set to 0.5, and they were all done on the testing dataset.

Keywords: Education across Lifespan, Input and Content Personalization Process, AI-generated outputs, Data Collection, Metrics.

1. INTRODUCTION

A big part of the fast growth of AI is Natural Language Processing (NLP) models and Large Language Models (LLMs). Anyone can use smart systems, even people who know little about AI or machine learning. At first, AI apps were simple because they came with tools like chat and translation apps. At the time, ChatGPT is often used to write things down during Q&A sessions [1]. This has made a lot of people interested in these tools now. Because of this, there are more AI tools for making things. They are simple to use and

help you get better at what you do. If you want to work tomorrow, you must know about AI today. It will change everything, and people worldwide must know how to use it. There are lots of ways to learn. You should teach them AI along with technology and the web. Some people do this to stay sharp at work. People worldwide want AI and how it can be

used in different fields, like the social sciences, to be a big part of college classes.

The school and the kids may need to change this add-on. Smart AI tools can help kids learn more when they use them. They can use what they've learned, work together, make things, and discuss them to do this [2]. Most people believe this way of teaching is based on constructivism because of this main reason. However, we still need to fix some things before we can reach our school goals. It all comes down to facts and what is right and wrong. It can be used to make movies, cartoons, music, songs, ideas for things and pictures, and more. Apps that turn text into pictures help people get things done faster and think of new ideas [26]. Some artists try to be real in their work and make it look more like them. Everyone should be able to keep their ideas and rights safe.

Kids should learn how to use AI tools by having schools teach them how. This is what people must do to talk to each other. AI can help us in many ways. However, we must be careful and follow the morals and rules already set to get the most out of it [3]. It is important to learn about AI because it helps people understand what it does and how it changes everything. These four things about AI are important to know. One of these is getting better at AI. AI is one way to get things done and ensure they're right. Using AI in different settings can also make many people think about what is right and wrong and how to figure that out [4]. AI has a lot of ways to change teens' lives. It can alter

how they learn, their chances of getting better jobs, and their knowledge of what people value. Schools need to be very careful about how they use AI these days. There are already very strong ways to use new AI in school. This is true for both students and teachers. The Department of Communication and Digital Media teaches it. Because AI is hard to teach in the social sciences, teachers tell their students to think of unique ways to learn the difference [27].

This is why there are now various ways to teach. That way, their theme-based ideas can come to life. They need to mix and find the best AI tools for making things. They were told to stick to the idea and find fun ways to teach AI through films [5]. In college, GenAI is helpful because it can help with one-on-one training, writing computer code, reviewing papers, and developing writing and thinking tasks. It can also help you make science units and grade sheets. They were asked to rate how they felt about using GenAI to help teach and learn at two different types of schools. One group was students. Many people who have used it said it was better than a search engine. Also, they said it's likely that kids will use GenAI tools even if their schools don't teach them how to [6]. A study found that chatGPT use in college makes teachers and students feel differently. Lozano and Blanco-Fontao's study found that the pupils liked GenAI methods used in schools and colleges. What they told you about ChatGPT didn't make them think it was dangerous.

This is the main reason why schools are getting worse. Sanchez-Ruiz et al. say that 110 students were asked to fill out a poll about ChatGPT and what they thought were its pros and cons [7]. The kids liked ChatGPT, as shown by the lines. Still, some people worry about how they'll be able to solve problems without using their normal thinking. The Sustainable Development Plan of the UN says that everyone should be able to go to school and do well. There are still places that lack the tools needed to meet this goal. There are times when GAI can help and hurt things in the classroom [8]. GAI should be used in all schools around the world. This is more true now that COVID-19 isn't getting bigger. People used to teach in ways that made many kids fall behind in maths and reading. People in this field are interested in GAI because it can make new things,

even though things change quickly [9]. Technology can do many things, such as change how people learn and assist an area in growing. People want to know how it affects the growth of kids in school.

Many people have talked about GAI's morals and the digital gap, even though it could help many people learn and meet many needs. Schools don't always adapt to each child's needs. Kids might get mad and not want to work hard after this. With GAIs, everyone could learn better, and school would be more fun [10]. Still, one issue that keeps coming up is that places that aren't making a lot of money don't have enough good tools for teaching. GAI can change this by giving everyone more ways to learn. Everyone will get to school faster.

Things could improve more if more people joined GAI [11]. People could also use AI-powered machine language systems to learn about other countries and join learning groups worldwide [28]. In school, boys and girls should have the same chances. This would help groups that don't always get a voice. GAI gives teachers more time to do important things, like teaching and one-on-one lessons. Be smart about using GAI, and don't let holes get bigger. Raman looked at how AI is used in schools to find the best way to use it for writing [12].

People relying too much on GAI might not work on important skills like thinking critically, getting better, making friends, etc. Don't let the digital gap get bigger. Make sure that everyone uses GAI tools in the same way. Everyone with a stake in GAI needs to work together to make it follow SDG4's ideas and goals. Rules should explain how to make, use, and accept GAI, among other things. Not only that, but teachers should also keep learning so they can use GAI well in their lessons.

2. Literature Review

Some people in the UK set rules for how big schools should use GenAI tools. AI can help kids if more people understand and learn how to use it. GenAI should be used to teach and grade; everyone should be told to collaborate and share their best ideas. Don't lie or cheat when you use Google's AI at school. Be good, smart, and honest instead. Gen AI and AI shouldn't be used in some ways in

schools. The Educational, Scientific, and Cultural Organization (UNESCO) made them. It's hard for people to talk about things because of UNESCO rules [13]. Also, they are meant to make AI simple for everyone. UNESCO can now make laws, rules, and standards to help people use GenAI technologies better in school, research, and study. Chan's study aimed to get people to think of an "AI Ecological Education Framework." People from several Hong Kong schools, including workers, students, and teachers, filled out the poll.

They want to change how schools teach and how kids learn with AI. In this part, AI can teach, test, and learn. We look at things like duty, privacy, and safety in the government area [29]. At the business level, you teach, help, and keep track of things. During the study, the groups made rules for how to use GenAI tools. These three people read the rules. There is a list of the fifty best schools in the world in TheWORLDUniversityRanking for 2023. According to experts, many of those schools' websites have rules about AI. According to their study, rules are "need of the hour" to keep things clean. Giving teachers the tech they need to talk to their kids about is one way to do this. Everyone who reviewed it agreed that teachers should also learn to use GenAI tools [14]. Farrelly and Baker found that more students from under-represented groups or international students whose first language is not English break the rules of academic ethics by using GenAI technologies than students from the United States do the same thing. So that GenAI tools could be used well in schools, they said that people should learn about AI and how to use it.

They also believed it was very important for the program to teach kids how to use GenAI fairly and smartly. Many people also said that teachers should learn about AI and how to talk to people who write AI programs so that schools can get great GenAI tools. They examined policy papers and thoughts from thirty of Shanghai's top 500 schools from May to July 2023. The study brought up a lot of important moral problems, such as duty, access, and having power over other people [15]. AI should be used in schools, but it's up to the teachers to figure out how. This would fix things when smart AI makes them worse. These two people looked at the GenAI rules that eight schools in Denmark made. Most of the rules they

learned in Danish schools were about using GenAI to do good tasks without breaking the law. But these rules didn't cover everything important. There was no talk of politics, business, the environment, or the long-term state of the climate. Also, how Google's AI changed how people learn wasn't discussed.

2.1. Education across Lifespan

Also, someone is always learning new things to improve their job or themselves. Tech, the job market, and work all change quickly, so everyone has to learn new things and get used to them. But this idea is very important. People learn all the time, not just after school. To always learn something new, think about this [16]. This kind of school teaches people how to be better people and workers. Everyone agrees you must keep learning to keep your job and adapt to new things. To learn more, everyone must go to school and follow the rules. It's said repeatedly in the European Commission: read and learn new things. Some say it makes you a better person, helps you fit in, and grows [30]. UNESCO's Education for Sustainable Development (ESD) program talks about how important it is to have ways of teaching that help everyone learn, from little kids to older adults. These events teach people information, skills, and habits that will help them grow.

2.2. AI's Impact on Education

More and more people believe that AI will change the way we learn. Everyone can learn in their way. Many more people will be able to go to school, and they can learn for the rest of their lives. It changes lots of things, like how people learn. Since they don't have to do dull, daily tasks, they have more time to do more important things, like teaching. AI can now learn smart ways, process data in real time, and teach itself smartly. This makes it easier for students to do their work [7]. That is, everyone can learn in ways that are easy for them to understand. Still, the study discusses the good and bad things that could happen when using AI in schools. But AI can help with school. AI can make school learning better, which is a good thing. All of these kids can be helped in this way. AI can be used to create a lot of different ways to learn. These will let you change the program so that everyone can learn in the best way for them. These tools allow you to monitor a student's growth, flaws, and

skills. When you talk to these kids this way, remember that they may have come from different places and had different amounts of life experience [31].

They can quickly figure out what they need more help with when they have trouble with a subject. Keeps them focused on what they need to do, which helps them learn more. AI is very important because it helps people connect what they learn at school with what they do outside of school.

2.2.1 Assessment in Tertiary Education

You should go over things and get comments when you're in college. They affect many people, including children, teachers, and people who work in management. Kids can do better in school, stay motivated, and learn to review better if you give them tests and notes. They can also teach kids new things. Many studies have been done on how tests affect how college students learn and how well they do in school. Other people can see your grades and notes. You can learn, teach, and run a business in different ways. They had to do well on these tests because meeting these work and school goals would help them improve in school [18]. People in college have to take a lot of different kinds of tests. These tests come in different forms: initial, first, final, and internet exams. People often look over their work with a group and also judge themselves. What works best for a subject depends on the school and what the kids need to learn [19]. These tests, also called "assessments for learning," are meant to help people learn. When you say "assessments of learning," tests should come after what was learned.

2.2.2 Evolution of AI in Education

AI (which stands for "artificial intelligence") changed a lot about how people learn when it was first thought of in 1956. If you use it wrong, it might not work right. But it gives kids answers that are more relevant to them, which might help them learn more. Some say having real teachers is important because computers can't judge or develop new ideas. This is still true, even though many think computers can teach instead of people [20]. Don't forget that AI shouldn't be used instead of school, and you should know how to use it right. Nethi and Moturu both said that schools should use AI more and more. These tools are designed to help

you learn more together, faster, better, and more effectively. This means that you are more likely to remember what you have learned. Remember this because using artificial intelligence (AI) in schools can help teachers do their jobs better. This is meant to show how important AI-powered e-learning tools are in changing how we learn.

2.2.3 Using AI in Student's Assessment

People say that COVID-19 has made it easy for people to learn online and on their own time. These are the first tech items that kids use at school. As AI has grown, high-tech tools that can make a big difference have become possible. This proves that GPT-3 can write long pieces just as well as those written by a person [21]. It can also make lesson plans and fix bugs in files. "Artificial intelligence" (AI) can mean many different things. But it does mean cool things can be done with tools. "Artificial intelligence" means that computers and apps can think and act like people. Being smart in this way means learning, guessing, and acting in ways that weren't planned ahead of time. A field of study that tries to make tools and computers do things independently is also called that.

2.2.4 The Potential of Gen AI in Assessment

Teachers need to know what their students have learned, so tests are a big part of the school. Giving them good input is simple when you know what they need to work on. Plus, it helps them do even better. The newest technological breakthroughs have changed how tests are given and how people teach and learn [22]. Tests that used to be on paper are often changed to ones that can be taken online. This is true because tests on the internet are more open, simple, and right for everyone. It's also clear from these tests that the best ways to test and grade have changed and improved over time.

2.2.5 The Factors Influencing the Usage of Gen AI in Students' Assessment

AI and other new tools could make schools better in many ways. One of the most important ways to do this is to give grades. However, there are some things that AI needs to improve on before it can be used in everyday life [23]. You should think about these things. Scores these days are mostly based on what Gen AI thinks. Gen AI excites many students because they think it will help them learn and write

better. Also, people only want to use Gen AI when they think it's worth more. However, students are afraid about their privacy, whether the tools are right, morals, personal growth, and the standards of society [24]. They need tools that use AI. The government and teachers must look at these parts if they want Gen AI tools to help students deal with their worries and fears and make tests safe. It's better to teach and learn in college in the long run.

3. Materials and Methods

The next part will have four subsections, each discussing a different part of the general study process. The first part discusses the methods and steps used to experiment. The second part discusses how different forms of the material were made with generative AI. The third part discusses how we got the data, and the fourth part examines how we did it.

3.1. Methods

People from a university in Europe who were in their first year of learning software engineering took part in the study in the second term. Forty-seven students participated in the study during the second term when the experiment happened [25]. No one was forced to take part in the study; everyone who did so did so voluntarily. Twenty of the 47 kids who signed up did the work. All the kids in this class are from Generation Z, born between 2001 and 2003. There were 15 guys and five women who took part. Their grade point scores at the end of the school year also showed that they

did well in many different grades. This picture shows how the study has changed many parts of school growth. The whole thing was about OOP, which stands for "object-oriented programming." The subject is split into two equal parts, each being looked at separately. They learned about the object-oriented computer model in the first part of their lessons. Someone learned everything they knew from an open-ended questions (OOP) book in six weeks. After six weeks of studying, part of the test was given.

For five more weeks, they did the second part of the study. This one taught object-oriented programming (OOP) and had another test to prepare students. They were given a book to follow along with and a study tool made by AI to help them in the second part of the course. The lessons in this guide can be explained in three different ways. Computer tools made all of them. The teacher would put the students' work into the learning management system (LMS) every day after class. On top of that, the teacher gives the lesson topic, the subject taught, and the work to be done. Once this information is given, it will link to the OpenAI API and make learning tools to help each student reach their learning goals. An English teacher made one of the first copies of this work. This type was the first one made. The students chose the other two. They were based on Batman and Wednesday Addams, two well-known icons from movies and TV shows. The kids were being watched as they switched from one type to another. The whole event is shown in Figure 1.

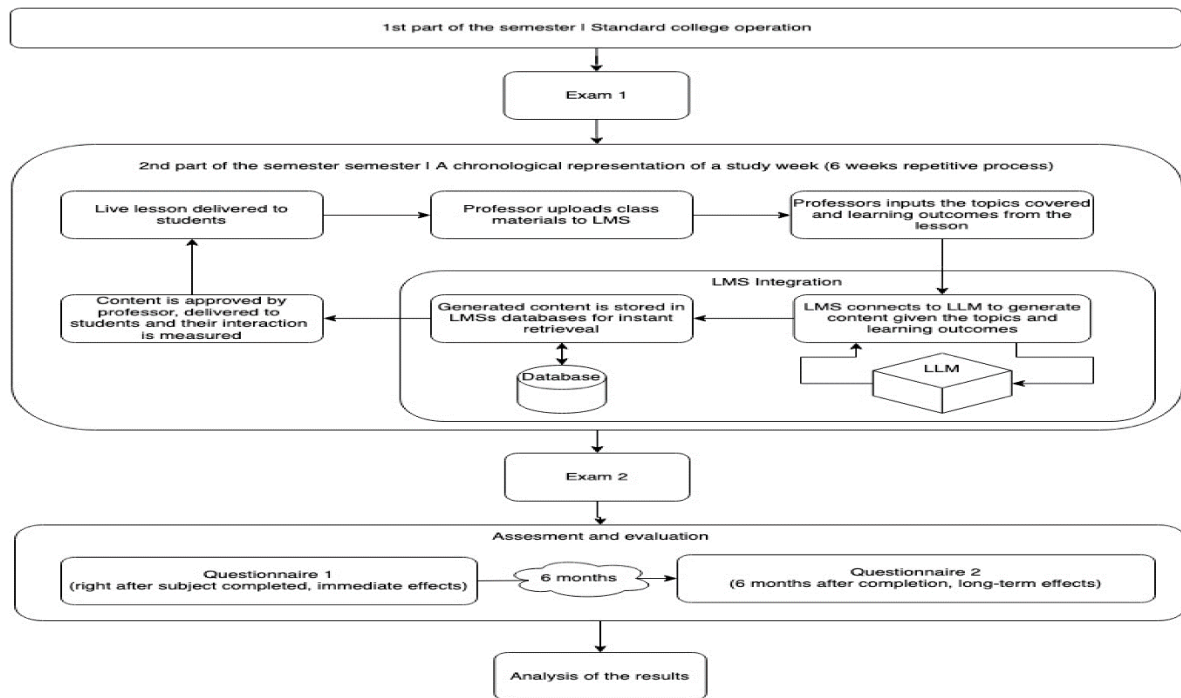


Figure 1. An example of the phases included in the experiment.

3.2. AI-Powered Content Creation and LMS Integration

The kids' learning management system (LMS) was changed so computers could make things independently. The GPT from OpenAI is a state-of-the-art language model that can make a wide range of learning tools. They had to study for a while, so they thought of making things that looked like people from pop culture. The kids could pick between a game with Wednesday Addams or Batman. Wonder Woman, Homer Simpson, James Bond, and Yoda were also brought up along with the ones already named. People don't just read about Bruce Wayne; he's a big part of the culture. This likely is why many people voted for him. Tuesday Night Lights got the second most votes. That's likely because Wednesday was a popular TV show on a big streaming service during the test. AA's third example of content was also added, which was also added that was made immediately. It sounds like a computer science teacher wrote it. Using OpenAI's API during the test was expensive, so only three were made.

This is more than one way to help kids learn. They were given three kinds of classwork to read and learn. These were some of the first things that were said. It helps both the kids and the adults.

After each class, the teacher made the work, checked it, and gave it to the students. In other words, it wasn't made because someone asked for it. We've already discussed how this method ensures the generation settings are right and that the teacher checks the work one last time for mistakes. To make things even better, this stops creative AI from hurting people in the first place. You can switch between them for fun, to try out new ways to teach, or to get a better grasp of the subject from different points of view. The next step of the method breaks the merger process into all of its steps and talks in great detail about each one.

3.2.1. Input and Content Personalization Process

Once teachers knew about the learning management system (LMS), they were given a new choice: the artificial intelligence content creator. The teachers had to fill out three sections that reflected the three most important parts of teaching: the title of the lesson, the topic or themes covered in the lesson, and the learning results. After the form was sent, the (LMS) made three API calls to the OpenAI API. Table 1 shows the settings for these calls.

TABLE 1. THE API FOR OPENAI REQUESTS INPUTS.

Part	A message from the computer
Shown	GPT-4
What's Up	It looks like "role" should be "system" and "content" should be "Static." It was shown in "job" = "user", "content" = "Title: [title]". That which it says: {content}. The learning goal is: {learning goals}. Next, add the following label (####) and two to four multiple-choice questions about the explanation that was made. Each question should have at least four possible answers. In JSON, the questions should be put like this: "question":{string}, "answers":[{"answer":{string}, "correct":{boolean}}]}
Temperature	6

Controls how unpredictable and unique the answers are. Undoubtedly, 0.7 will produce more random data while still being accurate. For each of the three separate API calls, a different system message told the GPT-4 model what part to play and how to finetune the outputs that were wanted from the model. Table 2 shows the system messages that were used.

TABLE 2. WITH EVERY API REQUEST, THE GENERATIVE AI MODEL RECEIVES SYSTEM NOTIFICATIONS.

Part	Message from the system
Teacher of computer science	People learn how to code from you. You always make things very clear by giving lots of examples. You're not sure that your kids understand what you mean. Don't start with an answer to a question; give more information.
Batman	Because you're Batman, everything you write has to have Batman talking. You always explain things very well by giving many examples. You're not sure if your kids get it. Don't answer the question right away; instead, give more details.
What Wednesday Addams	If you want to write like Wednesday Addams, you must show you are here. You always make things very clear by giving lots of examples. You're not sure that your kids understand what you mean. Don't start with an answer to a question; give more information.

The following information was entered, assuming that the teacher is teaching the kids about recursion from a computer point of view:

- **Title:**coming back;
- **Contents:**For example, use the PHP programming language to show what

recursion means in computer programming;

- **Learning outcomes:**Students should understand recursion, know when to use it, and be able to write basic functions that use it.

Figure 2 shows the OpenAI API call used to make the Batman version. You repeat the same steps twice more for the other two. There is only one difference: the system message is not shown, but the notes in Table 2 are.

```

1 {
2   "model": "gpt-4",
3   "messages": [
4     {
5       "role": "system",
6       "content": "You are Batman and everything you write must
sound like it is Batman talking. You always explain with great
details and with examples. You do not assume your students have
prior knowledge. Do not start with response to prompt, just start
explaining."
7     },
8     {
9       "role": "user",
10      "content": "Title: Recursion. Content: Explain what is
recursion in computer programming using the PHP programming language
for the examples. The learning objective is: Students should
understand the concept of recursion, should know when to use it and
should be able to write simple recursive functions.. After that put
the following separator #### and append between 2 and 4 multiple
choice questions with at least 4 possible answers about the
generated explanation. The questions should be JSON encoded in the
following format: {\\"questions\\": [{\\"question\\":
{string},\\"answers\\":[{\\"answer\\":{string},\\"correct\\":
{boolean}}]}]}"
11    }
12  ],
13  "temperature": 0.7
14 }

```

Figure 2. A system request was issued to the OpenAI API to create one version of the material.

3.2.2. AI-Generated Outputs

The OpenAI API always returned the exact type of data asked for when the system message was sent. Question types written in JSON ensure they will work with the LMS and be saved properly. This helps the students find the questions and show them in the best way possible. The answer sent in response to the question in Figure 2 can be seen in Figure 3.

In the shadows of computer programming, a concept lurks that's known as recursion. It's a method where a function, much like a vigilante, calls upon itself to solve a problem, breaking it down into smaller and more manageable tasks.

Let's take an example in the PHP programming language. Imagine you're in a labyrinth, much like the one I found myself in at Arkham Asylum. You're at a junction with multiple paths leading forward. The recursive function is like your guide, making a choice at each junction and proceeding down a path.

```

'''php
function labyrinth($path) {
    if ($path == 'exit') {
        return "Found the exit.";
    } else {
        return labyrinth($path + 1);
    }
}
'''

```

In this function, the base case is finding the exit (\$path == 'exit'). This is when the function stops calling itself. The recursive case is when the function calls itself, seeking the exit by advancing down the path (\$path + 1).

However, recursion isn't always the answer. Much like how I don't always use my Batmobile for every mission, you need to know when to use recursion. Recursion can be less efficient than iterative solutions in some cases, due to the overhead of repeated function calls. It can also lead to a stack overflow if the base case is not correctly defined, much like how Gotham would overflow with crime if I were not there to stop it.

To write a recursive function, you need to identify a base case that will end the recursion, and a recursive case that will break down your large problem into smaller ones. Think of it as a plan of action - every mission needs a clear objective and a strategy for achieving it.

```

#####
{"questions": [{"question": "What is recursion in computer programming?", "answers": [{"answer": "A function that calls other functions.", "correct": false}, {"answer": "A function that calls itself to solve a problem.", "correct": true}, {"answer": "A method to increase the speed of a program.", "correct": false}, {"answer": "A type of loop in PHP.", "correct": false}]}], {"question": "What can happen if the base case in a recursive function is not correctly defined?", "answers": [{"answer": "The program can crash due to a stack overflow.", "correct": true}, {"answer": "The program will execute faster.", "correct": false}, {"answer": "The function will not be able to call itself.", "correct": false}, {"answer": "It will convert the function into an iterative one.", "correct": false}]}], {"question": "What does a base case do in a recursive function?", "answers": [{"answer": "It defines the main problem to be solved.", "correct": false}, {"answer": "It provides a condition for the function to stop calling itself.", "correct": true}, {"answer": "It makes the function call other functions.", "correct": false}, {"answer": "It increases the efficiency of the function.", "correct": false}]}]}

```

Figure 3. Answers to a query from the OpenAI API are used to provide a Batman-style description of recursion.

Once the answer is obtained, the learning management system (LMS) is set up to add new questions and learning materials to its database. This means that the students can access them. The learning management system (LMS) divides the information into five parts, each with its name. Afterward, the first part is saved as the subject-related stuff. Then, each question and its answer are added to the learning management system (LMS) database. Before this can be done, the second part of the JSON must be cracked. The learning management system (LMS) uses data from the database to show class resources and quiz questions. This works whether the prompts and tools were made by hand or by computer. OpenAI's application programming interface (API) should be used to get the data, which should then be saved in the LMS network.

3.3. Data Collection

It was possible to see each student's grade, note, and how long they spent on each type of lesson. The learning management system (LMS) was also changed to track every student's use of a resource, not just AI-made resources. This was done so that these things could be tracked. Each time the person had the page open, a new call came in. This helps us figure out how often the kids talk to each other. Most computers today will think the tab is closed even if the kid doesn't close it. This means that

people won't always ask for things. The information would not be as correct if we only kept track of when someone closed a tab, though. It could stay open for a long time. They were told to use the course review box to let the teachers know if anything didn't make sense since AI made the whole thing. Everyone spent about the same amount of time on each subject type during each lesson. It also shows how many times and for how long people used each variant before and after the AI-generated content variant feature was added. The things the group that got their information from the LMS did are shown in Table 3.

TABLE 3. CHARACTERISTICS TAKEN FROM THE STUDENT'S LMS USE STATISTICS. BM, WA, AND CS ARE BATMAN, WEDNESDAY ADDAMS, AND COMPUTER SCIENTISTS.

A feature	An explanation
Lookup	The student's unique ID number
Rated by CS	The grade a certain student gave the computer scientist copy
Rated by WA	What grade does a certain student give the Wednesday Addams version
BM_Rating	The grade a certain student gave the Batman version
Check out CS Total Time	Total amount of time spent on the computer science version
Average time for CS	How long people usually spend on the computer science version
Total Time for WA	Total amount of time spent on Wednesday Addams version

Everyone who participated in the study got an evaluation soon after the second test and the end of the course. What did the kids think of this new way to make training tools and give them to other kids? That's what the study asked. Ten of them filled out this form and sent it back. After six months, we sent a new question to everyone who took the class to find out what long-term effects our suggested tool has on most of them. Thirteen of your peers filled out the second form and sent it back. Each poll has a different set of questions. They are in Table 4.

TABLE 4. QUESTIONS WERE SENT TO THE STUDENTS TO EVALUATE THEIR EXPERIENCE WITH THE PROPOSED INSTRUMENT IMMEDIATELY AFTER COURSE COMPLETION AND SIX MONTHS LATER.

Question	Sort	Survey
Please rate how well you felt about the training information you were given.	5-point Howrt	First, once the subject was finished
How well do you think you understood what was taught in each class?	5-point Howrt	First, once the subject was finished
The use of role-playing to teach made what I was learning better.	5-point Howrt	First, once the subject was finished
The ways that the lessons were taught in this class helped me remember them better.	5-point Howrt	First, once the subject was finished
This is how other kids should be taught.	Yes/No	First, once the subject was finished
What would you choose for all your classes for the rest of your studies if you had to?	Pick several options	First, once the subject was finished
Do you still remember an idea or lesson from class? Please explain.	Open-ended	Second, after 6 months
In what ways do you think the new role-based plan worked better or worse than the old ones?	Unfinished	Second, after 6 months

3.4. Data Analysis

We didn't tell anyone what the kids taught us until it was looked at. This was done so their grades wouldn't appear when the kids were put into groups. Someone was kicked out of class because they only used the learning management system (LMS) about three standard deviations of the time that everyone else did. Because of this, 19 people had to be looked at more closely. The study looked at how many points differed between the two tests and how well the students did on each one. Also, time was taken away from each kid. The study aimed to find a good reason to connect the two. With only 19 people, the group is too small to be useful for studying. They didn't use some statistical tests because they didn't know enough about them.

First, they were split into two groups based on how long they had used the app. By adding up how many times each student logged in, you could see how long they spent on each type of work. The group was found by adding up all of these numbers. To be "more active," a student had to do things longer than most peers.

They spent more time on technology than most other kids in their group. Some kids used the site about the same amount or less than their friends. These kids were called "less active." People called kids who spent the same amount of time with other kids or less than usual "less active." Keep track of how long students work on a job. This is a popular way for schools to guess how well students will do in school. An AI tool like K-means or a decision tree model could also be used. That being said, they can help with cluster analysis, but not enough to be useful. We can then put the kids into groups based on their busy schedules. People change what they do and how the tool works over time because they use it differently. Last, you can check out small sets side by side. A bigger study used the results of both polls. Questionnaires with many questions are a good way to find out what people know. In the first form, this kind of question came up most often. A score of 4 or 5 made the person very happy. After getting a score of 1, they were very upset.

They were mostly happy when they got a score of 2 or 3. Most of the time, this seven-point system worked best. However, only three measures were left after cutting the number from five to three. Different people have ideas about how to set up the polls for the next time when there are more children. The main point of the study was to look for patterns in these groups, especially connections between being very busy and being very happy. The quizzes, strange pictures, and questions that made people think were fun. On the second form, most of the answers were blank, and the answers were filled in by hand in these groups. Based on what the students said, these groups were made. There is room for everyone in the class. People's choices and the best ideas were put down and looked at when time allowed.

4. RESULTS

It's been cut in half, giving you two different pieces. In the first part, the writers say that these are the levels that the model reached on the test sample. In the second part, they talk about what happened and give many ideas for more study. They also compare their method to the standard method in the third part of the study. In the fourth piece, the last part of the study, the writers talk about how the research has been used in the real world.

4.1. Metrics

Table 5 shows the micro-, macro-, and weighted total F1 values for each class. To find the micro- and macro-averaged F1 scores, the authors used the map with the dataset to change the real and predicted mood labels to emotion labels. Table 7 shows how much memory and accuracy each class has [32]. It also shows how well each mood group class did regarding accuracy and memory.

TABLE 5. THE ACCURACY, RECALL, AND F1-SCORE METRICS FOR EACH CLASS ATTAINED ON THE TESTING DATASET ARE WEIGHTED, MICRO-, AND MACRO-AVERAGED (SOURCE: AUTHOR'S WORK).

Accuracy	Remember	Weighted F1-Score	F1-Score (Micro)	F1-Score (Macro)
61322	55658	56341	58353	50704

TABLE 6. WEIGHTED METRICS FOR ACCURACY, RECALL, AND F1-SCORE ARE AVAILABLE FOR EVERY CLASS ATTAINED ON THE TESTING DATASET (AFTER CONVERSION) (SOURCE: AUTHOR'S WORK).

F1-Score (Micro)	F1-Score (Macro)
7760	7349

TABLE 7. THE TESTING DATASET WAS USED TO DETERMINE EACH EMOTION CATEGORY'S ACCURACY AND RECALL SCORES (SOURCE: AUTHOR'S WORK).

Feelings	Accuracy	Remember
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Feelings	Accuracy	Remember
admiration	68873	73473
having fun	76567	88365
anger	47297	52239
a bother	31169	32432
approval	49838	39386
giving	51351	42222

The graphs for both the micro- and macro-averaged ROC curves are shown in Figure 4. It's 0.936 times the area under the macro-averaged curve but only 0.9361 times the micro-averaged curve. On top of that, Appendix B has receiver operating characteristic (ROC) curves. For each mood class, the area under each curve is found.

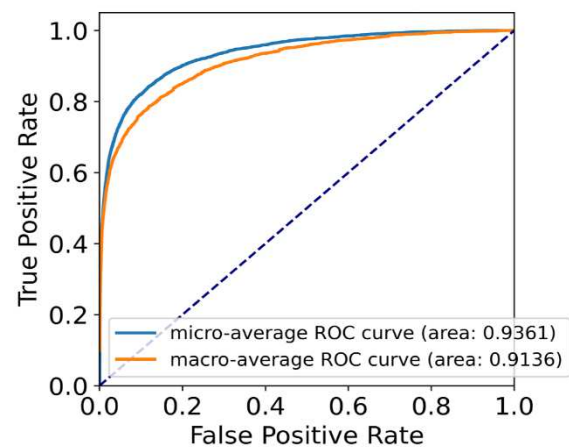


Figure 4. Micro- and macro-averaged roc curves (source: author's development).

You could also test the filter by making a grid that is hard to understand. There are charts in Appendix C that are hard to understand for all kinds of feelings. Let's talk about Appendix D, which has the tough-to-understand all-class grid. This was done with the all-vs-all method. There should be a list of all the steps to figure it out. This matrix can be made for only one type of multiclass classifier since each member only needs to match one expected class to one true class. It's not possible to make any other plans. The people who wrote the review copy ensured that each word had only one meaning. This is how the plan came together. It

was always chosen the class whose number was most likely to be correct.

4.2. Discussion

First, Table 6 shows that the model gets it more than 99% of the time and has a memory of more than 0 for each mood group. This is something that all the groups can say. It was used in a class with an AUC and memory of 0. There's not enough information for it to work well. Some groups were hard to understand or remember, and the collection has already been called unfair. It worked well after adding too many groups with bad feelings to the training set. The main tools used in the study were the micro- and macro-averaged F1 scores. This was done to see how the different kinds compare. Many people liked the macro-averaged score because it treats all groups equally and works well with unfair data. You can see that the all-vs.-all confused grid in Appendix D has a different line of right replies. It can tell the difference between and guess different types of feelings, as shown by the last result and this one. The model can mess up classes that should be linked smartly. This needs to be brought up when we talk about the plan.

Why do people not understand each other when disgusted, angry, or amazed? We don't understand them. When the all-versus-all confused matrix is used, this is what takes place. Use words that sound a lot like what someone is saying to show how you feel. You should use many strong words to show anger and irritation. However, you should use many question forms to show interest and unsure what to say. There is no doubt that this promise was kept because the F1-Scores are higher on both a small and a large scale. Good at telling the difference between good, bad, or not-so-clear feelings. This supports the idea that the model can tell if the thoughts shown are good, bad, or unclear. It's possible that the model can't tell the difference between two feelings that mean the same thing. We still don't know much about feelings because this is how we group them. It can be used for questions in many different areas of study, not just data science. These could be from any field that studies how people act. It might be useful to find out how people feel and what they think about what we learn in government or business.

There are people in the crowd that you want to talk to. It can guess how people will feel based on how they answer text polls. This is one way to find out things. Also a big part of the process. We got a lot of information about this group of feelings, and the guess worked well when there was a link between the two. This is a common way to measure things for people who work with machine learning, and this study's results back that up. Under Class D, you can find this all-vs.-all maps. This example shows how the model could guess more accurately about love, laughter, praise, and respect after getting more information. We can see this for ourselves. Shame, relief, happiness, and sadness were better not being able to explain than being able to name them. A smaller group of people talk about each of these thoughts. When fewer parts are in a class, the model can't learn and test as much. It's important to remember that, so let's talk about it. We can pick up on different kinds of thoughts. You can see that these are the receiver operating characteristic (ROC) forms for those feelings.

Some groups, like sadness, happiness, pride, and fear, don't have a lot of test data points. The way they look makes them look tired. This is true because the forms do show when someone is tired. As you can see in the third picture in Appendix A, the test information was put into groups. BERT's first study shows that making the model bigger makes it better at GLUE tasks. Someone who talks better has more levels, secret units, and attention heads. A mood indicator was found in this study. It might work better if taught with a more complicated BERT model, like BERT-large. One study also found that learning slowly helps you do better. This is an extra point on top of what was said. But remember that you'll need more computer tools to show these two concepts. Say this over and over. More data to train and test the classification should make it work better. This is because the model is still being taught and checked out. You can reach this goal by getting a new collection or adding to the one you already have.

4.3. Comparison with a Baseline Approach

In the part called "6. Transfer Learning Experiments," the researchers who wrote the study on the GoEmotions dataset looked to see if it could be used to make models. The BERT-base model

that had already been trained had 12 layers and 768 secret units. They showed what the normal finetuning had become. Based on the basic model, it got a score of 46 for guessing feelings and a score of 69 for guessing mood. They have a more complicated model that does a better job of pre-processing text and getting datasets ready. This is how the people who wrote it talk about it. It took 50704 points for the writers to get how people would feel right and 7349 points to get what people would think right. They could guess how people would feel 10.2% better and what they would think 6.5% better after that. This study's dataset has a different break between the train and test steps.

This is important to remember. At the end of the day, this is not only the best way to classify data but also the best way to work with new data. The most important thing that can be done to improve things is to add an extra text standards layer and more model choices. This will improve the trial collection. The first data set from GoEmotions was used to train and test the model. There were no changes to the splits. After that, it would be simple to match it with other types. Though it is being used now, the writers think the neutral class shouldn't be used in this study. In Table 8, you can see the results of the tests used to figure out mood.

TABLE 8. USING THE ORIGINAL GOEMOTIONS DATA IN EMOTIONS PREDICTION, THE AUTHOR'S WORK OFFERS THE F1-SCORE MICRO- AND MACRO-AVERAGED METRICS FOR EACH CLASS ACHIEVED ON THE TESTING DATASET.

F1-Score (Micro)	F1-Score (Macro)
4684	3442

"Grief," "nervousness," "pride," and "relief" are words that the model either doesn't remember or gets wrong. This study does not have as many of these effects as in the first one. This can be shown with the help of a tiny BERT model with two layers and 128 secret units. This makes sense since computers don't have a lot of power. It should work better to sort if a BERT model with more bits is used, like a BERT base. The BERT-base model is where GoEmotions begins its study to go along with what was already said. The splits are one more

thing to think about. This will help the model find and tell the difference between all the different types of feelings. To do this, split the dataset again by class and add more low-class data. In other words, eliminating the neutral class makes the model work better. Their study shows that "if raters weren't sure about any emotion being expressed, they were asked to choose Neutral." Raters could pick "no emotions" if they weren't sure how to put a sample into a group. They could tell us what they needed. No matter what case it was, it was thrown out.

This method can be used in two ways: words can be put down or not. There shouldn't be any mood groups turned on by level if no feelings are shown. The biggest group, the middle class, is also to blame because the plan works too well for everyone. Because of this, the different groups are not a part of this probe. As you may know, the study only recommended teaching the simple tiny-BERT model, which has two layers and 128 secret units. Nothing special was added to the given bits; the text had not been read first. Hyperparameters were used the same way before, even though they needed to be changed. Also, there was no middle class. Taking some steps made it possible to guess how someone would feel.

TABLE 9. THE F1-SCORE MICRO- AND MACRO-AVERAGED METRICS FOR EACH CLASS WERE OBTAINED ON THE TESTING DATASET FOR THE LIMITED MODEL IN EMOTIONS PREDICTION (SOURCE: AUTHOR'S DESIGN).

F1-Score (Micro)	F1-Score (Macro)
5648	4909

The numbers here don't match up with the ones in the measures. Averaging the F1-Score scores shows that adding more model layers and text pre-processing makes the model 3.27 percent better at guessing how people feel and 4.11 percent better at guessing how people feel about something. The main result is seen when a mismatch is solved and the neutral class is thrown out. It was trained on the first split of GoEmotions, but the neutral class was not included. This was done to make sure of this even more. These numbers came up in the tests done to determine the mood.

TABLE 10. F1-SCORE MICRO- AND MACRO-AVERAGED METRICS FOR EVERY CLASS WERE OBTAINED FOR THE TESTING DATASET USING THE ORIGINAL GOEMOTIONS DATA IN EMOTIONS PREDICTION, EXCLUDING THE NEUTRAL CLASS (SOURCE: AUTHOR'S INVENTION).

F1-Score (Micro)	F1-Score (Macro)
5677	609

This shows that the F1-Score worldwide is 9% less than before the various splits were used. Appendix E shows that the model is better at guessing feelings than "grief." It does, however, often guess "sadness" or "fear" in a real "grief" class. These groups don't like it. Remember that the all-versus-all plan might not be the best choice. There are things in the testing part of the first GoEmotions split that have more than one name. The expected class is the one whose score is most likely correct. This is the reason why these things happen. As for "grief," "nervousness," and "relief," the model wasn't good at remembering or being right. So, it can't tell the difference between these groups. Some models, like GoEmotions, only work when you use a method that unfairly splits data. This kind of care is needed. There are many split algorithms, but they all need a test case to show how well they work. A useful number in this set of data shows "unseen" information. TeachingTeaching, validating, and measuring thoughts and moods in separate groups has worked better so far. This is because the neutral group is removed, the text is more regular, and the classifier's structure gets harder.

4.4. Practical Implications

They talk about a tool that can figure out how people are feeling by reading what they write. It can read English text messages written in a language many people use in chat apps, so you don't need to do anything else. This model can be used to find out how people in a group feel. The people who write can also use their thoughts to improve things in the future. You can try out different BERT models and make changes to get a better rank. If you go to <https://github.com/Shoomaher/sentiment-analysis>, you will find all the codes used in this project.

5. LIMITATIONS

Several things should be changed about this work. It lacks enough info to learn, so it can't do so. Texts in any other language are unknown to GoEmotions because it only has English texts. It's also true of how the work is put together. It's possible for very different writing styles not to work well together, even if they make the text more fair, like writing a long blog post about using the method instead of a short message. Also, you should know that the model can only read things that are written down. To find out how people feel in this study, neither words nor body language are used. The people who work at GoEmotions also said that most text comments come from Reddit, which young guys mostly use. In other words, the model might be unable to read books written for very different groups. Third, some sets of facts aren't fair. It can't say that because it doesn't know enough about these groups. We've already talked about this.

The model also has another flaw: it can only guess a certain number of thoughts and feelings. It depends on the data used to build the model. More research is also needed before it can correctly guess other mood groups. It needs to be taught with fresh information first. Ultimately, they only had enough data and computer power to train and test the model a few times. A lot of people use big language models these days. They're useful for many things and should be better at grouping feelings. Most of the time, these models need a lot of data and computer power to learn.

6. CONCLUSIONS

Deep learning will be used to find out how words make people feel for this project. Models can now use deep learning to look through huge amounts of data and find difficult trends. The GLUE test, which checks how well the machine can understand spoken words, is another proof. The finetuning method for the pre-trained BERT model was used to train and research how to guess how people will feel. People talked about the links using this set from GoEmotions. This work has more information about a text-standardization layer. Cleans up the model's data to handle input as raw text. Even though standardizing text was being shown, it can also be used in other models that take text from English chat rooms. We did well when we used the

F1-Score measure and macro-averaging. This is why the plan works well. It was 10.2% better at guessing people's feelings and 6.5% better at guessing what they thought than the usual method. It was also good at guessing all the mood-type groups in the dataset, even though some groups didn't have a lot of data.

Declaration:

All authors declare that they have no conflicts of interest.

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